

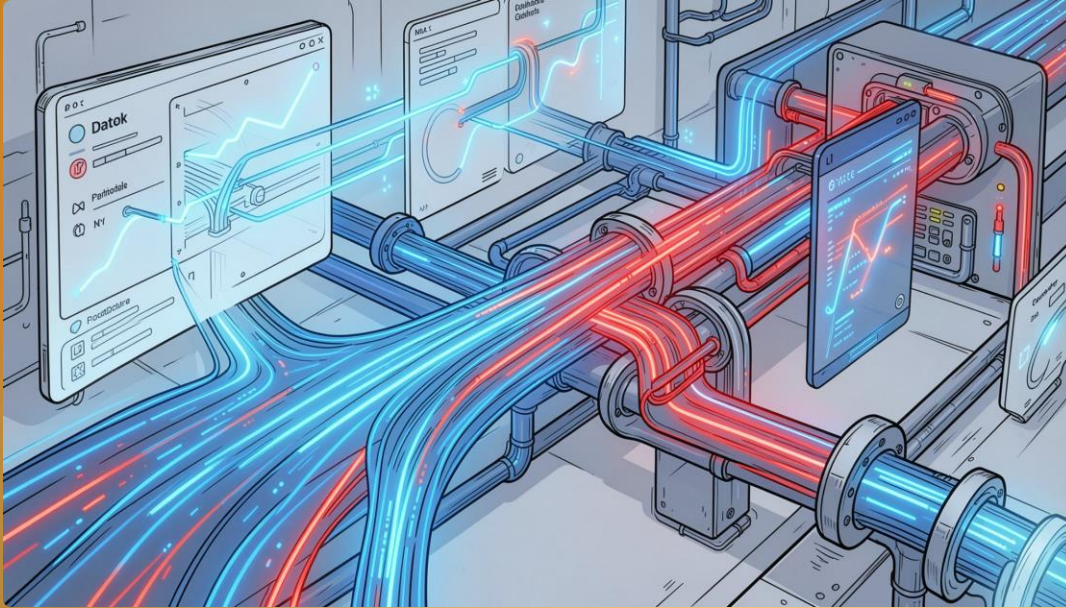


What Is Power Query?

Your essential guide to transforming, cleaning, and shaping data — from raw and messy to analysis-ready. Built into Excel and Power BI.

POWER QUERY FUNDAMENTALS

Power Query: The Data Preparation Engine



Power Query is a **data connection and transformation tool** built into Excel and Power BI. It lets you import, clean, reshape, and combine data — without writing a single line of code.

Extract

Import from files, databases, web, and more

Transform

Clean, filter, pivot, and reshape your data

Load

Send the result to Excel or a Power BI model

Why Power Query Changes Everything

⚡ Repeatable

Steps are recorded automatically. Refresh your data with one click — all transformations replay.

🔑 No Code Required

A visual interface handles the heavy lifting. Advanced users can edit the underlying M code.

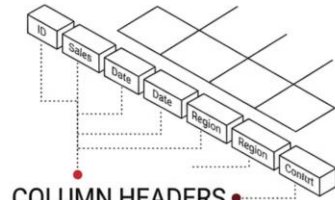
👁 Unified Workflow

Works the same way in Excel and Power BI. Skills transfer directly between tools.

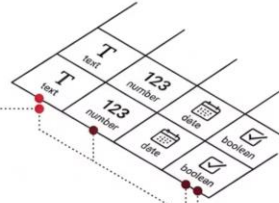
📊 Audit Trail

Every transformation is logged in Applied Steps — easy to review, edit, or undo.

Key Table Concepts

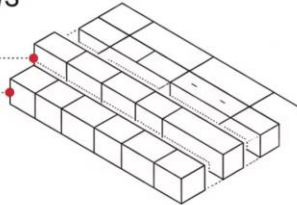


COLUMN HEADERS
NAMES THAT IDENTIFY EACH FIELD



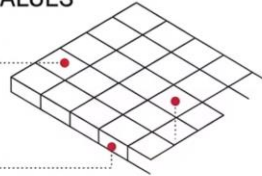
DATA TYPES
MUST BE SET CORRECTLY

ROWS



EACH ROW IS ONE COMPLETE RECORD

NULL VALUES



EMPTY CELLS THAT MUST BE HANDLED BEFORE ANALYSIS

Why Structure Matters

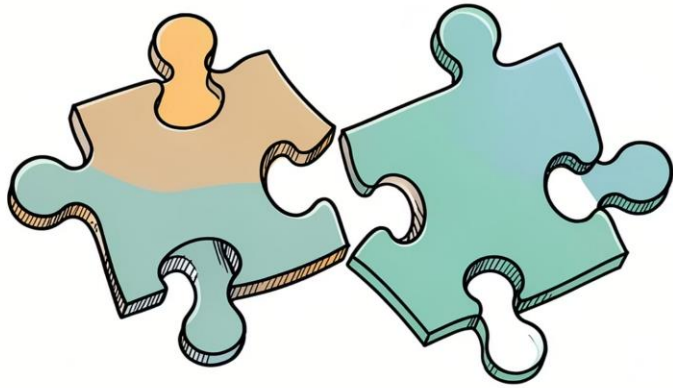
Power Query relies on consistent table structure to apply transformations correctly. Misaligned headers, mixed data types, or blank rows can break your entire query.



Pro Tip: Always use "Use First Row as Headers" after importing raw data to establish clean column names.

Merge Tables: Joining Data Sources

Merge combines two tables side-by-side by matching rows on a **key column** — similar to VLOOKUP or SQL JOIN.



1

Left Table

Your primary dataset (e.g., Orders)

2

Join Key

Matching column (e.g., CustomerID)

3

Right Table

Lookup table (e.g., Customers)

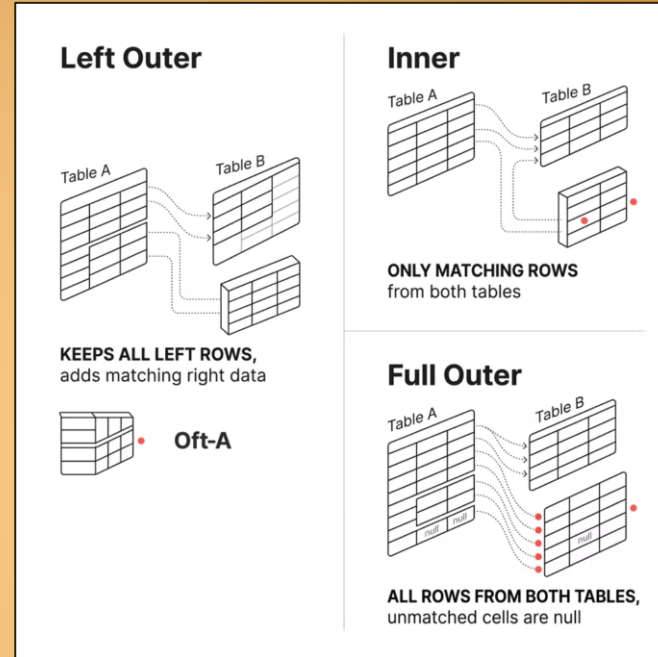
4

Merged Result

Enriched table with combined columns

Merge Join Types

Choosing the right join type determines which rows appear in your result. Power Query supports six join kinds:



Left Outer

All rows from left table + matches from right

Inner

Only rows that match in both tables

Full Outer

All rows from both tables

Append Tables: Stacking Data

Append adds rows from one table to another — ideal for combining **monthly reports**, **regional files**, or **partitioned datasets** into a single table.

Table A

Jan–Mar Sales
500 rows

+

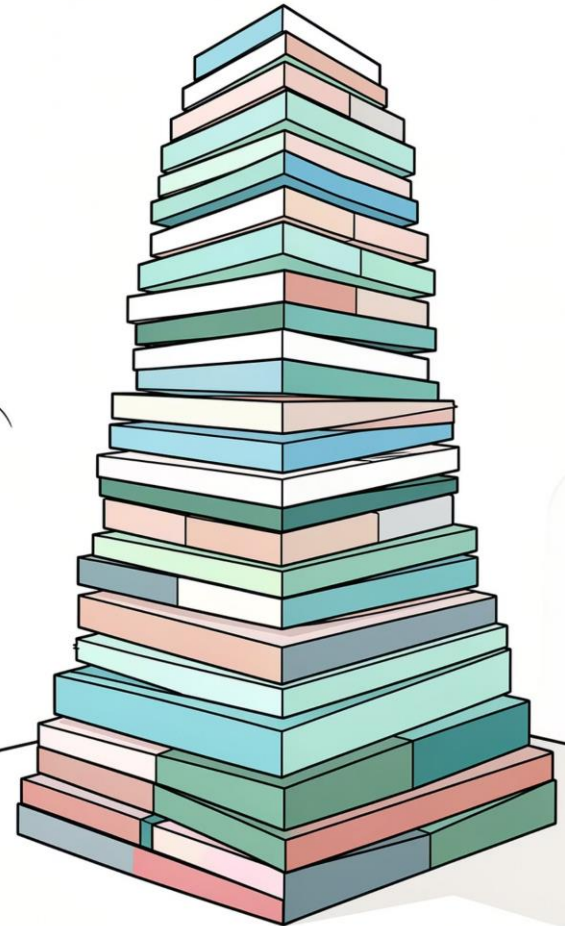
Append

Table B

Apr–Jun Sales
500 rows

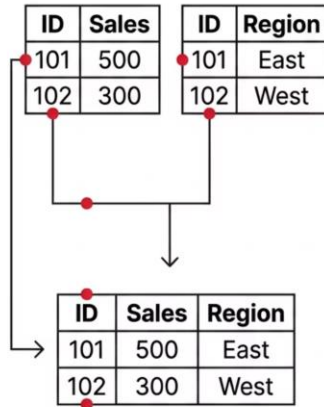
= Result

Full Year Sales
1,000 rows



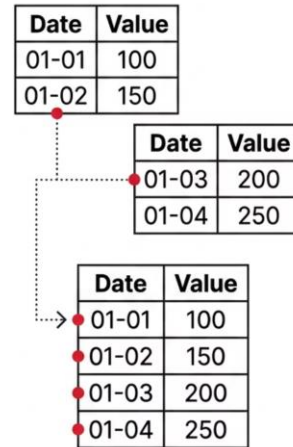
Merge vs. Append: When to Use Which

Merge



Adds columns by matching rows on a key (horizontal expansion, like VLOOKUP)

Append



Adds rows by stacking tables (vertical expansion, like UNION)

Quick Decision Guide

Use Merge when...

You need to bring in additional columns from a related table using a shared key field.

Use Append when...

You have the same columns across multiple tables and want to stack them into one.



Watch out: Appending tables with mismatched column names creates nulls. Standardize headers first.

Merge vs. Append: When to Use Which

MERGE (Combine Tables by Matching Columns)

Merge adds columns from one table to another based on a related column (like Excel VLOOKUP)

Table 1: Customers		
CustomerID	CustomerName	City
101	John	New York
102	Sara	Los Angeles
103	David	Chicago
104	Emma	Houston

Table 2: Orders			
OrderID	CustomerID	OrderDate	Amount
1001	101	01-Jan-24	500
1002	102	02-Jan-24	700
1003	101	03-Jan-24	300
1004	103	04-Jan-24	400
1005	105	05-Jan-24	600

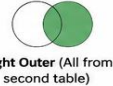


Result After Merge (Matched on CustomerID)					
CustomerID	CustomerName	City	OrderID	OrderDate	Amount
101	John	New York	1001	01-Jan-24	500
102	Sara	Los Angeles	1002	02-Jan-24	700
101	John	New York	1003	03-Jan-24	300
103	David	Chicago	1004	04-Jan-24	400

- Combines tables horizontally (adds columns).
- Uses a matching column (key).
- Similar to SQL JOIN / VLOOKUP.



Left Outer (All from first table)



Right Outer (All from second table)



Inner Join (Matching only)



Full Outer (All records)



Left Anti (Only non-matching from first)

Common Join Types in Merge

APPEND (Combine Tables by Adding Rows)

Append adds rows from one table to another. Both tables must have the same columns.

Table 1: Sales_Jan			
Date	Product	Salesperson	Amount
01-Jan-24	Laptop	John	1200
02-Jan-24	Mouse	Sara	150
03-Jan-24	Keyboard	David	200

Table 2: Sales_Feb			
Date	Product	Salesperson	Amount
01-Feb-24	Laptop	John	1100
02-Feb-24	Mouse	Sara	180
03-Feb-24	Monitor	Emma	300



Result After Append			
Date	Product	Salesperson	Amount
01-Jan-24	Laptop	John	1200
02-Jan-24	Mouse	Sara	150
03-Jan-24	Keyboard	David	200
01-Feb-24	Laptop	John	1100
02-Feb-24	Mouse	Sara	180
03-Feb-24	Monitor	Emma	300

- Combines tables vertically (adds rows).
- Both tables should have same columns.
- Similar to SQL UNION ALL.

MERGE vs APPEND

MERGE

- Combines by matching column
- Adds columns
- Horizontal combination
- SQL JOIN / VLOOKUP

APPEND

- Combines by adding rows
- No matching column needed
- Vertical combination
- SQL UNION ALL

When to Use?

- ☒ Use **MERGE** when you want to add more information (columns) from another table.
- ☒ Use **APPEND** when you want to combine similar data from multiple sources (rows).

Example:

- Monthly sales files → APPEND
- Sales + Customer details → MERGE



Thank You learners..
Let's explore **Power Query**.